**ABERDARE TECHNICAL COLLEGE**

**INFORMATION COMMUNICATION TECHNOLOGY**

**MODULE 2,TERM 2**

**UNIT:COMPUTER REPAIR AND MAINTAINANCE**

**UNIT CODE:0714 351 04A**

**ELEMENT 3:TEST COMPUTER COMPONENT FUNCTIONALITY**

**Total marks: 30 | Time allowed: 6 hours**

**Candidate details**

Name: _____ Reg. No: _____ Date: _____

Computer/Station No: _____

**INSTRUCTIONS TO CANDIDATES**

- ✓ *Write your name, registration number, date, and station number in the spaces provided on the front page before you begin.*
- ✓ *Attempt **all five tasks** in the order given. The paper is marked out of **30 marks** and you have **2 hours**.*
- ✓ *Back up any files on the USB flash drive before you start, because the drive will be completely erased. Double-check the drive letter and size in imageUSB so you do not erase the wrong disk.*
- ✓ *Do not switch off, unplug, or restart the computer while MemTest86 is running, unless the supervisor instructs you to do so.*
- ✓ *Record your observations on the answer sheet as you work (version, RAM size, passes, errors, and result). Answers written after the test from memory will not be accepted.*
- ✓ *Do not change any BIOS/UEFI settings except those needed to boot from the USB, and note any change you make. Raise your hand and ask the supervisor if you have a problem with the equipment or the download.*

**Requirements**

- A computer to test (desktop or laptop) with at least one RAM module installed
- A second working computer **with internet access**, used to download and prepare the USB
- A USB flash drive of at least 1 GB. **Everything on it will be erased, so back up any files first.**
- Pen and the answer sheet at the end

**TASK 1: Preparation and download**

1. On the working computer, open a web browser and go to the official site: **https://www.memtest86.com/download.htm**
2. Download the USB image directly using this link: **https://www.memtest86.com/downloads/memtest86-usb.zip** *(If the direct link fails, use the "Free Download" option for "Image for creating bootable USB Drive (Windows)" on the download page.)*
3. Right-click the downloaded ZIP file and choose **Extract All**. Open the extracted folder. It contains imageUSB.exe and the image file memtest86-usb.img.
4. Before testing, record the RAM information of the computer to be tested. In Windows, press **Ctrl + Shift + Esc**, open **Performance > Memory**, and note the total size, speed, and slots used.

**TASK 2: Creating the bootable MemTest86 flash drive.**

1. Insert the USB flash drive into the working computer.
2. Run **imageUSB.exe** as Administrator (right-click > *Run as administrator*).
3. In **Step 1**, select **only your flash drive**. Check the drive letter and size carefully so you do not select another disk.
4. In **Step 2**, choose **Write image to UFD**.
5. In **Step 3**, click **Browse** and select memtest86-usb.img (it should already be selected).
6. Click **Write**, confirm the warning, and wait until you see the message that the write is complete.
7. Safely eject the flash drive.

*Alternative tool: **Rufus** (https://rufus.ie) can write the same image in DD Image mode.*

**TASK 3: Booting from the flash drive.**

1. Switch off the computer to be tested and insert the prepared flash drive.
2. Switch it on and immediately press the boot menu key repeatedly. Common keys are **F12** (Dell, Lenovo), **F9** (HP), **F10**, **Esc**, or **F8** (ASUS).
3. Select the **USB flash drive** from the boot menu and press **Enter**.

4. If the computer will not boot from the USB, enter BIOS/UEFI setup (usually **F2** or **Del**) and temporarily disable **Secure Boot** or enable USB boot. Record any change you make.
5. Wait for MemTest86 to load and show the main screen.

**TASK 4: Running the memory test**

1. On the main screen, press **Enter** or click **Start Test** (some versions show a configuration screen first, so keep the default settings).
2. Observe the information MemTest86 displays and record it on your answer sheet: RAM size, speed, and type (DDR3, DDR4, etc.).
3. Let the test run. Note the **test number** being run, the **pass number**, and the **elapsed time**.
4. Allow **at least 1 full pass** to complete. If time allows, complete 4 passes, which is the default.
5. Watch for the **Errors** counter. If any errors appear, write down the number of errors and the test during which they occurred.
6. Do not switch off or touch the computer during the test unless supervised.

**TASK 5: Results and report**

1. When the test finishes, a results screen shows **PASS** or **FAIL**. Record it.
2. Press **Esc** (or click Exit) to leave MemTest86, then remove the flash drive and restart the computer normally.
3. If the version used saves reports, a report file is stored on the USB in the EFI\BOOT folder (HTML/log). Insert the USB into a working computer and view it.

***The trainer is introducing oral questions shortly.***